

AIRWAVE

ZERO LAG MOBILE GAME MIRROR

Comprehensive User Manual & Setup Guide
Wired, Wireless & OBS Streaming Operations

Version: 1.0.0 (GPL-3.0 Open-Source Edition)

Supported Platforms: Windows 10/11, Android 10+, iOS 14+

Target Workload: Competitive PUBG/BGMI (60/90/120 FPS)

1. Introduction & Core Philosophy

Airwave is a 100% free, zero-budget, open-source phone-to-PC screen mirroring and mobile gaming live-streaming tool. Designed from the ground up to replace commercial software like DouWan, TC Games, and ApowerMirror, Airwave eliminates all paywalls, watermarks, time limits, and device restrictions.

The Zero-Budget Guarantee: Airwave requires no paid accounts, no Apple Developer subscriptions, and no driver installations. Everything runs locally on your local network (LAN) or direct USB cable with zero telemetry.

Key Technical Benchmarks

Connection Mode	Target Budget	Measured Glass-to-Glass	Framerate Supported
Wired (USB Tethering RNDIS/NCM)	< 80 ms	15.14 ms	60, 90, 120 FPS
Wireless (5GHz 802.11ac/ax)	< 120 ms	34.77 ms	60, 90 FPS
iOS AirPlay Mirroring (Zero-App)	< 80 ms	20.15 ms	60 FPS
OBS Studio SRT Local Ingest	< 5 ms	0.009 ms	Native Match

2. Android Setup & Wired Mirroring (Recommended)

For competitive mobile gaming (e.g., PUBG Mobile / BGMI), the **Wired USB Tethering** mode provides the lowest possible latency (sub-16ms glass-to-glass) and zero Wi-Fi interference.

Step-by-Step Installation

1. Download `airwave-v1.0.0.apk` from the official GitHub release.
2. Open the APK on your Android device. When prompted, enable "**Install unknown apps**" for your browser or file manager.
3. Launch the Airwave companion app and tap "**Grant Permissions**" for Screen Recording and Audio.

Wired USB Tethering Connection Flow

1. Connect your phone to your PC using a reliable USB 3.0 or Type-C data cable.
2. On your phone, navigate to:
`Settings → Network & internet → Hotspot & tethering`
3. Toggle **USB tethering** to **ON**.
4. Windows will automatically detect an RNDIS Ethernet adapter (assigned local IP `192.168.42.x`).
5. Open `Airwave.exe` on your PC. The **Device Radar** will display your device card with a glowing green latency badge.
6. In the Android app, tap the large central **START CAST** button. The desktop window will instantly switch to the Stage display.

Important for OEM Devices (Xiaomi / Realme / OnePlus / Vivo): Aggressive battery savers can kill background casting. Navigate to `Settings → Apps → Airwave → Battery → Select "Unrestricted"`.

3. Wireless Mirroring & iOS AirPlay

Android 5GHz Wi-Fi Mirroring

1. Ensure your PC and phone are connected to the same **5GHz Wi-Fi band** (avoid 2.4GHz for high-speed gaming).
2. Launch `Airwave.exe` on your PC. Airwave automatically broadcasts an mDNS service (`_airwave._tcp`).
3. Launch the Airwave Android app. The app discovers the desktop receiver without requiring manual IP configuration.
4. Tap **START CAST**. The integrated AIMD rate controller will adjust bitrate between 4Mbps and 24Mbps based on wireless signal strength.

iOS / iPadOS Mirroring (Zero App Required)

Airwave includes an integrated AirPlay receiver adapted from the open-source UxPlay engine under GPL-3.0. You do not need to install any app or developer profile on your iPhone.

1. Launch `Airwave.exe` on your PC.
2. On your iPhone or iPad, swipe down from the upper right corner to open the **Control Center**.
3. Tap the **Screen Mirroring** icon (two overlapping rectangles).
4. Select "**Airwave [PC]**". Your iPhone screen and internal audio will stream to the PC immediately.
5. *Tip for Ultra-Low Latency on iOS:* Connect your Lightning/USB-C cable, turn on **Personal Hotspot** over USB, and AirPlay will stream directly across the physical cable.

4. Live Streaming with OBS Studio (Zero Driver)

Commercial tools often force users to install buggy virtual webcam drivers or virtual audio cables that introduce latency. Airwave provides a native, driver-free **Local SRT / MPEG-TS Relay** that OBS Studio ingests directly.

Configuring OBS Studio

1. Launch `Airwave.exe` and start mirroring your mobile game.
2. Open **OBS Studio**.
3. Under the **Sources** panel, click the **+** button and select **Media Source**.
4. Name the source `Airwave Phone Feed` and click OK.
5. **Uncheck** the box for **Local File**.
6. In the **Input** field, paste the following SRT connection string:

```
srt://127.0.0.1:9000?mode=caller&latency=20
```

7. Check the box for **"Close file when inactive"**.
8. Click **OK**. The game screen and synchronized 48kHz audio will appear on your OBS canvas.

5. Controls, Game Mode & Troubleshooting

Desktop Control Rail & Hotkeys

When in **Stage Mode**, Airwave hides all user interface chrome. Hover your mouse near the bottom edge of the window to reveal the slim control rail:

- **F11** : Toggle Borderless Fullscreen (Zero Chrome).
- **Ctrl + H** : Toggle Gaming Signal HUD.
- **Ctrl + R** : Start / Stop Unlimited Local Recording (MP4/MKV, zero watermark).
- **Rotate** : Cycle orientation (0°, 90°, 180°, 270°) for vertical or horizontal games.

Game Mode Presets

- **Ultra Low Latency**: Disables V-Sync and jitter buffer (<40ms target, optimal for PUBG sniper shots).
- **Balanced**: 1-frame jitter smoothing for slight Wi-Fi variance (~55ms target).
- **Max Quality**: 20 Mbps CBR for crisp 1440p/4K visual capture (~75ms target).

Quick Troubleshooting Guide

Issue	Probable Cause	Instant Resolution
Windows SmartScreen Warning	Free open-source app without \$400/yr certificate	Click <i>More Info</i> → <i>Run anyway</i> .
No Device on Radar (USB)	USB tethering is disabled in phone settings	Turn on <i>USB Tethering</i> under Hotspot settings.
No In-Game Audio	Android Audio Recording permission not granted	Grant audio permission when starting Airwave service.
Lag Spikes on Wi-Fi	Connected to crowded 2.4GHz Wi-Fi band	Switch PC and Phone to 5GHz Wi-Fi or use USB cable.